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Multi-feature stock price prediction by LSTM networks based on VMD and TMFG

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Abstract

The stock market is characterized by its high nonlinearity and complexity, making traditional methods ineffective in capturing its nonlinear features and complex market dynamics. This paper proposes a novel stock price forecasting model—the Variational Mode Decomposition—Triangulated Maximally Filtered Graph—Long Short-Term Memory (VMD–TMFG–LSTM) combined model—aimed at improving prediction accuracy, stability, and computational efficiency. The proposed model first employs Variational Mode Decomposition (VMD) to decompose the stock price time series into multiple smooth intrinsic mode functions (IMFs), reducing data complexity and mitigating noise interference. Subsequently, the TMFG algorithm is utilized for feature selection, simplifying the input data and accelerating the iterative convergence process. Finally, the filtered features are modeled and predicted using a Long Short-Term Memory (LSTM) network. Experimental results demonstrate that the VMD–TMFG–LSTM model significantly outperforms AutoRegressive Integrated Moving Average (ARIMA), Neural Network (NN), Deep Neural Network (DNN), Convolutional Neural Network (CNN), as well as single LSTM, TMFG–LSTM, and VMD–LSTM models in forecasting the closing prices of multiple stocks. Specifically, for Shanghai International Airport Co., Ltd. (sh600009), the VMD–TMFG–LSTM model achieves a 69.76% reduction in Root Mean Squared Error (RMSE), a 71.41% reduction in Mean Absolute Error (MAE), a 46.28% reduction in runtime, and an improvement of 0.2184 in R-squared (R^2), indicating significantly higher prediction accuracy. In conclusion, the combined model proposed in this paper enhances the accuracy, efficiency, and stability of stock price prediction, providing a robust and efficient solution for forecasting stock market trends.

Keywords: VMD, TMFG, LSTM, Stock price prediction

Introductory

Stock forecasting refers to the prediction of future stock price trends by analysing historical data from the stock market. Because of its highly nonlinear and complex nature, the stock market is considered one of the most complex financial systems [1]. Stock forecasting is a method of predicting future price changes by analysing historical data from the stock market, which has great practical value and theoretical significance for investors and market analysts. Traditional statistical methods, such as moving averages, exponential smoothing [2] and ARIMA models [3], although excellent in handling linear data, often struggle to capture the non-linear features and complex market dynamics

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in the stock market. With the rapid development of artificial intelligence technology, especially the advancement of deep learning, stock prediction methods have been innovated and revolutionised. Long Short-Term Memory (LSTM), a specialised Recurrent Neural Network (RNN) [4], has become a powerful tool for processing time series data. LSTM effectively solves the problem of gradient vanishing encountered by traditional RNNs in the processing of long time series data through its original gating mechanism, which enables LSTM to identify and exploit long-term dependencies in the data, greatly enhancing the accuracy of the model for time series prediction.

Nonetheless, the high volatility, nonlinearity, chaos, and operational complexity of the stock market itself make accurately predicting stock prices still a great challenge. In order to effectively address these challenges, this study proposes a state-of-the-art combinatorial model (VMD–TMFG–LSTM) that incorporates variational modal decomposition (VMD), Triangulated Maximally Filtered Graph (TMFG), and LSTM network. This model starts with an accurate decomposition of stock closing price series using VMD technique to reduce the complexity of the data and to effectively remove unstructured noise. Immediately after that, the processed data is sparsely dimensionalised by the TMFG algorithm, a step that not only reduces the burden of data processing, but also highlights the core features that influence stock price movements. Finally, the filtered key features are fed into the LSTM network for deep training to achieve accurate prediction of stock prices.

By combining these methods, the present model not only significantly improves the understanding of stock market dynamics, but also greatly enhances the practicality and stability of the model through its high adaptability to new data and strong prediction ability. The innovation of this paper is to combine the TMFG algorithm with the variational modal decomposition (VMD) and LSTM in order to solve the problem of stock price prediction with high volatility, nonlinearity, and chaos, and to enhance the prediction accuracy and efficiency. In addition, the development of this model provides new research directions and solutions in the field of financial technology.

The innovations and contributions of this study are as follows:

(1) The stock technical indicators are calculated based on the stock price data, and 43 technical indicators are additionally calculated, and the indicators reach 59 characteristics together with the 16 indicators already in the original data. The indicator system for stock price prediction is improved.
(2) The closing price is subjected to VMD modal decomposition. The VMD algorithm decomposes the stock price time series into multiple smooth IMF components, which reduces some of the test error noise interference and improves the prediction accuracy.
(3) The introduction of the TMFG algorithm to select the input features reduces the complexity of the input, improves the speed of iterative convergence and reduces the iterative error, and improves the stability and robustness of the prediction.

The rest of the paper is organised as follows, the second part is the literature review section, which provides an overview of existing stock prediction models and methods, discusses data decomposition and feature selection techniques, and introduces the

application of TMFG algorithm to filtered data and the application of LSTM to stock prediction. The third part is the Methods and Data section, which details the principles of VMD decomposition, the TMFG algorithm and the LSTM network, and describes the construction of the combined VMD–TMFG–LSTM model. It also describes the data sources and data preprocessing for this study. The fourth part is the experimental and result analysis section, which demonstrates the experimental process and results of VMD decomposition, TMFG processing and LSTM prediction, and compares the prediction effect of the models. The fifth part is the conclusion section, which will summarise the research results and model advantages, and discuss the limitations of the model and future research directions.

Literature review

Forecasting techniques for stock data are usually classified into two main categories: traditional and non-traditional statistical methods. Among traditional models, Autoregressive Moving Average (ARIMA) [5] and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) [6] are commonly used tools, typically based on assumptions of normal distribution and linear relationships between variables. While traditional statistical methods have a long-standing theoretical foundation, and their assumptions and properties have been well-studied and validated, they assume that the data are linear and normally distributed. However, actual financial time series data are often nonlinear and exhibit thick-tailed distributions, making it difficult for these models to capture complex patterns and long-range dependencies. This limitation may result in suboptimal model performance when dealing with complex nonlinear relationships. Non-traditional statistical methods primarily include machine learning and deep learning techniques. Algorithms like Support Vector Machines (SVM) [7] and Gradient Boosting (Boost) [8] have shown good results in stock price prediction. These methods are better suited for capturing complex nonlinear relationships, particularly in financial time series data, which often exhibit such characteristics. As technology advances, especially in the area of neural networks, deep neural networks have gained widespread attention in the analysis and prediction of stock price fluctuations. Models such as Convolutional Neural Networks (CNN) [9] and Long Short-Term Memory (LSTM) [10] have become particularly prominent. Researchers have developed numerous improved algorithms and optimization strategies to enhance prediction accuracy, achieving significant results in practical applications. These innovations not only advance financial forecasting technology but also provide a more scientific foundation for investment decision-making.

As a typical temporal prediction network in deep learning, the LSTM neural network has performed well in the field of stock price forecasting. For instance, Sun et al. [10] used an LSTM network to extract temporal features from data and predict the Shanghai Stock Exchange (000001), yielding better results than traditional statistical prediction methods. Similarly, Gao [11] conducted a study using a deep LSTM network to predict stock prices from six U.S. market stocks across five different industries, and the results demonstrated that the LSTM network had more stable generalization ability. Chen and Ge [12] introduced an attention mechanism into the LSTM model, forming the AttLSTM model, which significantly improved prediction performance for the Hong Kong stock market and further enhanced prediction accuracy. Tian et al. [13] proposed

a hybrid LSTM network model based on Pearson's correlation coefficient and Bayesian optimization of LightGBM to predict stock price fluctuations, showing better approximation and generalization capabilities than several other models.

Building on the use of LSTM networks for stock forecasting, scholars have also introduced data decomposition techniques, such as Empirical Mode Decomposition (EMD) and Variational Mode Decomposition (VMD). Srijiranon et al. [14] proposed a hybrid forecasting model (PCA-EMD-LSTM), which combines Principal Component Analysis (PCA), EMD, and LSTM to further improve the prediction accuracy of stock closing prices in the Thai stock market. However, this method is prone to mode aliasing. Niu et al. [15] constructed a hybrid VMD–LSTM model for time-varying series forecasting, demonstrating that VMD–LSTM can better fit time-varying data compared to EMD and single LSTM networks.

In addition to data decomposition, researchers have addressed the challenges of high volatility, nonlinearity, and chaos in the stock market by focusing on feature selection and data dimensionality reduction. Li et al. [16] applied the Least Absolute Shrinkage and Selection Operator (LASSO) to reduce the dimensionality of basic trading, valuation, and technical indicators, and then combined the processed data with an LSTM network for prediction. Their results showed that the LSTM model, after dimensionality reduction, achieved superior prediction accuracy. Similarly, Chen et al. [17] proposed a multi-feature stock price prediction model based on multi-feature computation, LASSO feature selection, and Ca-LSTM networks, considering the importance of each feature. Their experiments demonstrated that this model outperformed traditional time-series prediction models and standard LSTM methods in terms of both accuracy and speed.

To address the impact of high noise interference and short-term fluctuations on stock trend prediction, Wu et al. proposed a hybrid method based on discrete wavelet transform (DWT) denoising and extreme learning machine (ELM) [18]. By denoising stock data, this method significantly improved the smoothness and trend characteristics of the data, and outperformed 12 machine learning algorithms, including recurrent neural networks (RNN) and long short-term memory (LSTM), in predicting 400 Chinese stocks. Additionally, the study of nonlinear and complex fluctuations in financial markets has led to the exploration of stochastic volatility models. Research has shown that stochastic volatility models, combined with fractional differential operators and Fourier transforms, can effectively capture the dynamic changes in asset prices [19]. By comparing the performance of stochastic volatility models with classical models, such as the CGMY and Black–Scholes (BS) models, the study validated their effectiveness in predicting volatility and asset price trends under different time fractional orders.

For complex datasets, information filtering networks constructed from correlation matrices can retrieve meaningful structures and extract relevant interactions from multivariate datasets in complex systems. Recently, a new information filtering network, the Triangulated Maximally Filtered Graph (TMFG) [20, 21], was introduced. It is computationally faster and more suitable for large datasets [22]. Briola and Aste [23] study utilized Minimum Spanning Tree (MST) and Triangulated Maximally Filtered Graph (TMFG) methods to analyze the logarithmic return cross-correlations of 25 cryptocurrencies traded on the FTX exchange across different time resolutions (ranging from 15 s to 1 day), uncovering the evolution of dynamic hierarchical structures over varying time

scales. The study demonstrated that TMFG effectively captures key dependency structures across time resolutions with stability, statistical significance, and economic relevance. This provides significant support for the VMD–TMFG–LSTM model in stock prediction, particularly in feature selection in stock markets. Similarly, Hussain Shahzad et al. [24] applied the TMFG algorithm to construct an energy stock return dependence network using data from 105 global energy companies across 26 countries in the Americas, Asia–Pacific, and Europe. The extracted significant dependencies facilitated the development of a portfolio trading strategy, which outperformed the traditional equal-weighted buy-and-hold approach. These findings support the applicability of the TMFG algorithm in analyzing complex stock data, enabling the identification of critical relationships and enhancing data-driven decision-making in financial markets. Barfuss et al. [25] demonstrated that TMFG is highly effective for generating sparse probabilistic models with high likelihood. This method is well-suited for handling cluster structures in complex data, especially for time series analysis in finance, such as stock return modeling. It is based on local and low-dimensional inversion, offering computational efficiency and statistical robustness, even in the inverse covariance estimation of high-dimensional, noisy, and short time series.

Methods and data

The overall process of stock price prediction based on the TMFG-VMD–LSTM model in this paper is shown in Fig. 1.

The overall process consists of five main components:

(1) Obtain the price-related information of the target stock, including its opening price, closing price, high price, low price, Changes, volume, turnover, Circulation market value, total market value, turnover rate, restoration price, P/E ratio, P/S ratio, P/C ratio, P/B ratio, closing price recovery price, and other 16 indicators;
(2) Calculating the technical indicators from the original stock data resulted in 43 stock technical indicators, which were merged into the original data to obtain new experimental data.
(3) To process the experimental data, in order to better retain the characteristics of the original data, we use the VMD variational modal decomposition technique to process the stock price data, decomposing the stock price signal into 8 smooth IMF components, together with the original 16 indicators as well as 43 stock technical indicators, resulting in 67 indicators.
(4) The obtained 67 indicators are input into the TMFG–LSTM combination model as a sample data set. The combined model consists of input layer, TMFG layer, LSTM layer, ReLU activation layer, and output layer. Due to the large number of input indicators, we first input the dataset into the TMFG layer, which is filtered by the information filtering network to obtain the precision matrix, and then filter the input indicators to downsize the raw data, a step that not only reduces the burden of data processing, but also highlights the core features that affect the movement of the stock price. Finally, the filtered key features are fed into the LSTM network for deep training to achieve accurate prediction of stock prices.

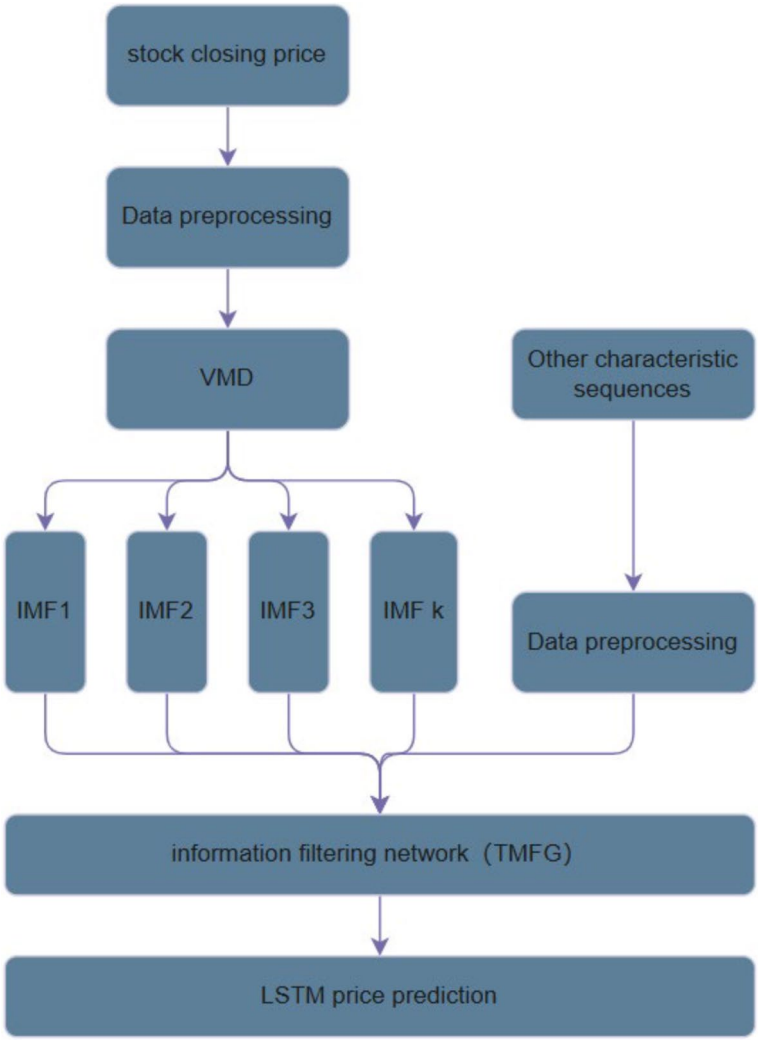


Fig. 1 Overall flowchart of the combined VMD–TMFG–LSTM algorithm

(5) To validate the model, another four stocks from SSE are taken and predicted under the same experimental conditions as the above experiments to verify the validity and robustness of this model.

VMD decomposition

Variational modal decomposition (VMD) is a nonlinear, nonsmooth signal analysis method for signal processing [26]. It decomposes a complex signal into a series of intrinsic modal functions (IMFs) through adaptive decomposition, where each IMF represents an oscillatory mode in the signal [27]. Compared with the classical empirical modal decomposition (EMD), VMD avoids the phenomenon of modal aliasing by introducing constraints, and VMD has better stability and accuracy in processing signals, which is especially suitable for dealing with multi-frequency signals.

The VMD decomposition is given by:

$$s_k(t) = \left(\delta(t) + \frac{j}{\pi t}\right) u_k(t) \tag{1}$$

$$\begin{cases} \min\limits_{\{u_k\}\{\omega_k\}} \sum_k \left\{ \left\| \mu_t \left[\left(\delta(t) + \frac{j}{\pi t}\right) u_k(t) \right] e^{-j\omega_k t} \right\| \right\} \\ \text{s.t. } \sum_k u_k(t) = y(t), \end{cases} \tag{2}$$

where $s_k(t)$ the analytic signal of the kth mode at time t, δ(t) is the Dirac function used to construct the spectrum of the mode, j is an imaginary unit, and $u_k(t)$ the kth eigenmode function component.μ_t is a control parameter used to adjust the smoothness of the decomposition, y(t) is the original signal, and ω_k is the centre frequency of the K modes.

TMFG algorithm

The TMFG algorithm (Triangulated Maximally Filtered Graph) is a method used to construct sparse networks [21]. It selects edges in the network through an iterative process, giving preference to those edges that maximise the triangulated graph. The algorithm first starts with the complete graph and then gradually removes edges until the desired sparsity level is reached. At each step, TMFG selects those edges that increase the number of triangles in the graph, thus ensuring the sparsity of the network while maintaining a high degree of structure. TMFG is used to deal with the structure of the clusters in graphical models, and is particularly suited to time-series analyses in the financial domain, e.g., modelling of stock returns. By calculating the maximal clusters, it is possible to identify the set of variables that are interrelated in a time series, which can then be used for risk management and portfolio optimisation. It can effectively reveal the complex interactions between variables. The TMFG algorithm can be used to analyse stock datasets by calculating the clusters structure to identify the strong correlations between variables and screen the features for stock price prediction. The 'sparsity' of the sparse inverse covariance matrix means that most of the elements of the matrix are zero, which is very useful in practical applications because it reflects only the key direct relationships between the variables and ignores indirect or non-significant relationships. This property makes the model easier to interpret and significantly reduces computational complexity and improves computational efficiency when dealing with high dimensional data.

The algorithm formula is as follows [25]:

In the second-order case, the model distribution that maximizes entropy while constraining moments for a given value is:

$$f(X) = \frac{1}{Z} exp\left(-\sum_{ij} \frac{1}{2} (X_i - \mu_i) J_{i,j} (X_j - \mu_j) \right) \tag{3}$$

where f(X) is the objective distribution function, Z is the normalisation factor, X_i, X_j are the variables i,j, μ_i, μ_j are the means of the variables i, j, and $J_{i,j}$ is the covariance matrix element between the variables i,j. The equation can be decomposed according to the relationship between the variables:

$$f(X) = \frac{\prod_{m=1}^{M_c} f_{C_m}(X_{C_m})}{\prod_{n=1}^{M_s} f_{S_n}(X_{S_n})^{k(S_n)-1}} \tag{4}$$

where $f_{C_m}(X_{C_m})$ is the local distribution of the clusters C_m, $f_{S_n}(X_{S_n})$ is the local distribution of the separations S_n, and k(S_n) is the normalisation constant of the separations S_n. Similarly, J can be decomposed, and we can obtain J from the summation of the local inverse of the correlation matrix.

$$J_{i,j} = \sum_{C s.t. \{i,j\} \in C} \left(\mathbf{\Sigma}_C^{-1}\right)_{i,j} - \sum_{S s.t. \{i,j\} \in S} (k(S)-1)(\mathbf{\Sigma}_S^{-1})_{i,j} \tag{5}$$

where $J_{i,j}$ is the covariance matrix element between variables i,j, $\mathbf{\Sigma}_C^{-1}$ is the inverse covariance matrix of regiment C, and $\mathbf{\Sigma}_S^{-1}$ is the inverse covariance matrix of separation S.

The process of constructing a filter network begins with initialisation: an initial simplex (tetrahedron) with four nodes is created, and the four nodes with the minimum correlation coefficient determinant are selected. The triangular faces and remaining nodes are then defined, and the four triangular faces of the initial simplex are assigned to the set T, and all nodes except the initial simplex are assigned to the set V.

Then select a triangular face (consisting of nodes {va, vb, vc}) from set T and a node vd from set V as illustrated in Fig. 2. Calculate the ratio of the determinant of the correlation coefficients when the node vd is added to the triangular faces to form the new quadruple simplex C ({va, vb, vc, vd}) and the new separator S ({va, vb, vc}). The node vd that makes the ratio of the determinant the largest is chosen so that the increment that maximises the logarithmic determinant is achieved. Then update the set: remove the new node vd from the set V, remove the old triangular faces {va, vb, vc} from the set T, and add to the set T the three new triangular faces formed by the new node vd: {va, vb, vd}, {va, vc, vd}, and {vb, vc, vd}. The non-zero elements of the inverse covariance matrix J are updated according to the new quadruple simplex C and separator S. End the iteration when the set V is empty, i.e., all nodes have been added to the network. Finally, the sparse inverse covariance matrix J is returned with non-zero elements indicating the existence of edges (direct relationships) between variables in the TMFG network and zero elements indicating that the two variables are conditionally independent.



Fig. 2 Schematic diagram of TMFG filtering network composition

LSTM network

LSTM is a kind of improved recurrent neural network, using LSTM can effectively transmit and express information in a longer time sequence without causing the effective information of a longer time ago to be forgotten [28, 29]. The cell module structure of LSTM network is shown in Fig. 3. It can be seen that the LSTM network uses memory cell records to transmit information, LSTM handles the time lag task through three control gates and uses the sigmoid function and the tanh function to update the cell state, the three gates are the forgetting gate, the input gate, and the output gate. The forgetting gate selects how much information from the previous stage can be retained to the current stage cell state, the input gate selects how much of the current stage input information can be retained to the current stage cell state, and the output gate selects how much of the current stage cell state is taken as the output value of the LSTM.

The formula for LSTM is:

$$f^{(t-1)} = \delta({}^{t}W_f[h^{(t-1)}, x^t] + b_f) \tag{6}$$

$$i^t = \sigma(W_i[h^{(t-1)}, x^t] + b_i) \tag{7}$$

$$c^t = \sigma(W_c[h^{(t-1)}, x^t] + b_c) \tag{8}$$

$$C^t = i^t * c^t + f^{(t)} * C^{t-1} \tag{9}$$

$$\begin{aligned} o^t &= \sigma(W_o[h^{(t-1)}, x^t] + b_o) \\ h^t &= o^t * \tanh^{-1}(c^t) \end{aligned} \tag{10}$$

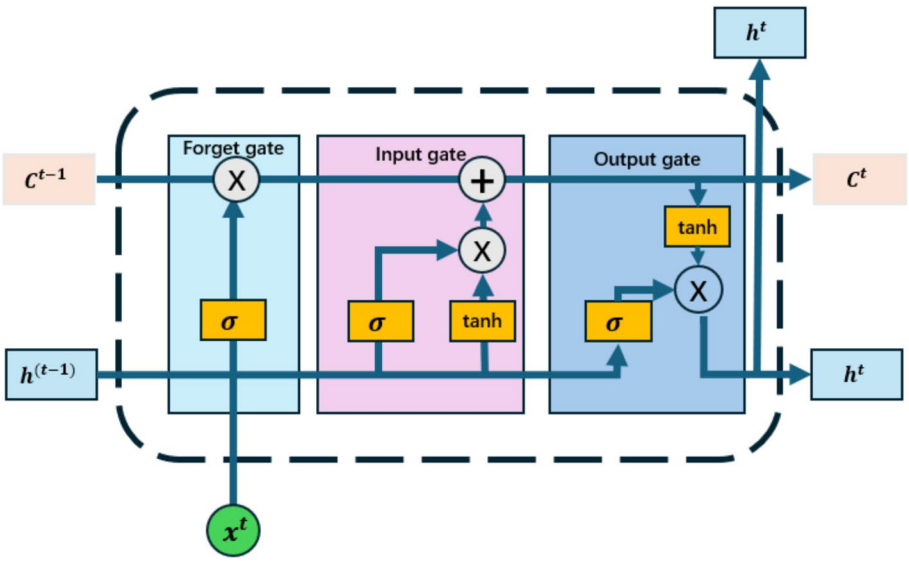


Fig. 3 Schematic diagram of LSTM network structure unit

where W_i, W_c, W_o represent the corresponding weights, b_i, b_c, b_o represent the corresponding biases, and c^t represents the current cell state value, and h^t is the output value of the current cell.

Evaluation indicators

In this paper, the commonly used method of evaluating regression prediction algorithms is used to evaluate the model effect, and four evaluation indexes, namely, Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination [R-Square (R^2)], and Model Running Time, are used to perform the model performance indexes Evaluation.

The Root Mean Square Error (RMSE) is calculated by the formula:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2} \tag{11}$$

The Mean Absolute Error (MAE) is calculated using the formula:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i| \tag{12}$$

The coefficient of determination is a statistical measure of how well a model predicts, reflecting how well the model explains the variability in the data. The value of R^2 usually ranges from 0 to 1. The closer the value is to 1, the better the predictive ability of the model and the more variability it explains.

The formula is:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y})^2} \tag{13}$$

where y_i is the actual value, $\widehat{y}_i$ is the predicted value, $\overline{y}$ is the average of the actual values, and n is the number of data points.

The model run time formula is used to measure the efficiency of the model run, by measuring the start and end time points we can get the length of time required to execute the process. In Python's model run, we use the time module to record the time before and after the model run and then calculate the difference. The formula for its calculation is:

$$T = t_{\mathrm{end}} - t_{\mathrm{start}} \tag{14}$$

where T denotes the total running time of the model, t_{start} is the timestamp when the model starts running, and t_{end} is the timestamp when the model ends running.

Data sources and pre-processing

Data sources

The original dataset of this paper's research comes from yucezhe (https://yucezhe.com/), and the Shanghai International Airport Co., Ltd. (sh600009) stock of Shanghai Stock Exchange is selected as the experimental stock, and Shanghai Airport (sh600009)'s

opening price, closing price, high price, low price, Changes, volume, turnover, Circulation market value, total market value, turnover rate, restoration pric, P/E ratio, P/S ratio, P/C ratio, P/B ratio, closing price recovery price are the original data. The original data indicators are shown in Table 1.

When constructing predictive models, the use of multiple price-related variables provides a comprehensive view of the market. Although the primary focus in this paper is on the closing price for prediction, incorporating additional price-related variables such as opening price, high price, low price, and trading volume enhances the model's ability to capture market conditions. These variables complement the closing price by providing insights into volatility and price movement trends, ultimately leading to more robust and accurate predictive capabilities.

Outlier handling

For the stock data of the opening price, closing price, high price, low price and up and down and other 16 indicators, from the forecasters network database to obtain the indicator data for descriptive statistics, found that the data have a small number of missing values, relative to the total number of 6,000 pieces of data accounted for a small proportion of the missing value is also scattered in the data did not focus on the emergence of the missing value, so using the method of the nearest neighbor to take the nearest to the missing data sample before and after each of the five data Therefore, the nearest neighbor method is used to take the five data before and after the nearest sample of missing data to fill in the missing data by weighted average.

Stock price technical indicators

Stock technical indicators are a tool used to analyse stock price movements and market sentiment and they are calculated based on historical price and volume data [30]. These indicators can provide clues about the future movement of stock prices, and although they do not guarantee accurate predictions, they still have an important role in stock analysis [31]. In practice, technical indicators are used as features input into a machine learning model, which learns the historical patterns of these technical indicators to predict future stock price movements, in addition, the use of technical indicators can help overcome the limitations of traditional fundamental analysis as they are not influenced by the fundamental factors of the company, but rather focus on the stock price

Table 1 16 Raw data indicators

Norm	Notation	Norm	Notation
Opening price	Open	total market value	market_value
Closing price	Close	turnover rate	turnover
Highest price	High	restoration price	adjust_price
Lowest price	Low	P/E ratio	PE_TTM
Changes	change	P/S ratio	PS_TTM
Volume	volume	P/C ratio	PC_TTM
Turnover	Money	P/B ratio	pb
Circulation market value	traded_market_value	Previous closing price recovery price	adjust_price_f

movement itself. When selecting technical indicators, it is necessary to consider their relevance, stability and diversity with respect to stock price movements. By carefully selecting and processing these technical indicators, powerful predictive models can be constructed to provide strong support for stock market analyses and decision-making. In this paper, we have selected several commonly used stock price technical indicators. The specific technical specifications are shown in Table 2.

In the selection of indicator parameters, we adopted standard technical indicator parameters and did not extensively optimize them. However, these parameters are independent of the features selected by TMFG. TMFG analyzes and selects the features most closely related to the prediction target (closing price) through its graph structure, which further helps us build a more accurate LSTM model.

Data normalization

In the modeling process, in order to eliminate the influence of the scale between the data and to solve the problem of high correlation between the variables as well as to improve the running speed of the model, the original data were linearly transformed,

Table 2 43 Stock technical indicators

Norm	Notation	Norm	Notation
10-day average	MA_10	10-day rate of change	ROC_10
10-day exponential average	EMA_10	14-day average amplitude indicator	ATR_14
20-day exponential average	EMA_20	20-day Bollinger Band	BollingerB_20
10-day moving average	SMA_10	20-day Bollinger Band %b	Bollinger%b_20
20-day moving average	SMA_20	turning point	PP
10-day momentum	Momentum_10	first resistance level	R1
first support point,	S1	second resistance level	R2
second support point	S2	third resistance level	R3
third support point	S3	the 14-day stochastic oscillator	SO%d_14
the 15-day triple exponential smoothing average	Trix_15	the 14-day average directional movement index	ADX_14_14
the 12-day fast and 26-day slow MACD	MACD_12_26	the 12-day fast and 26-day slow MACD signal line	MACDsign_12_26
12-day fast and 26-day slow MACD bars	MACDdiff_12_26	Mace line (high-low price trend reversal)	Mass Index
14-day vortex indicator	Vortex_14	KST oscillator	KST_10_15_20_30_10_10_10_15
14-Day Relative Strength Indicator	RSI_14	true strength indicator	TSI_25_13
Chaikin oscillator	Chaikin	14-day money flow and ratio indicators	MFI_14
14-day energy tide indicator	OBV_14	14-day Strength Index Indicator	Force_14
14-day simple volatility indicator	EoM_14	20-day trend indicator	CCI_20
Ten-day wave estimation indicator	Copp_10	Keltner Channel Center Line	KelChM_10
Keltner Pass upper track	KelChU_10	Keltner Pass lower track	KelChD_10
ultimate oscillator	Ultimate_Osc	Donchian Channels	Donchian_14
KDJ indicator-K	K	KDJ indicator-D	D
KDJ indicator-J	J		

and the method was chosen to be min–max standardized, and all the index data values were mapped to the interval [0, 1], and the transformation function was:

$$x^* = \frac{x - min}{max - min} \tag{15}$$

where max is the maximum value of the sample data, min is the minimum value of the sample data, and x is the original value to be normalized.

Experiments and analysis of results

VMD variational modal decomposition

VMD (Variational Mode Decomposition) is a signal processing technique that is commonly used to decompose time series data in order to extract its intrinsic patterns and trends [32, 33]. Compared with traditional smoothing techniques, VMD better preserves the characteristics of the original data during the decomposition process and therefore is more likely to capture important information and trends in stock price data. We finally set the parameters of VMD decomposition as follows: alpha (bandwidth constraint) is set to 7000, tau (noise tolerance) is set to 0, the DC value is 0, the init (initialisation mode, 1 means uniform initialisation) value is 1, the tol value is 1e−7 (This is a tolerance parameter used to determine the convergence criteria of the algorithm. When the improvement between successive iterations is less than this tolerance, the algorithm considers that it has reached convergence and thus stops iterating).

Before the VMD decomposition, the number of IMF eigenmode functions to be decomposed needs to be determined, i.e., the value of k needs to be determined. We compare the center frequencies of the VMD decomposition based on taking different values of k from 3 to 15. According to the center frequency plots of different values of k in Fig. 4 and the correlation coefficient plots of the IMF with the original signal in Fig. 5,

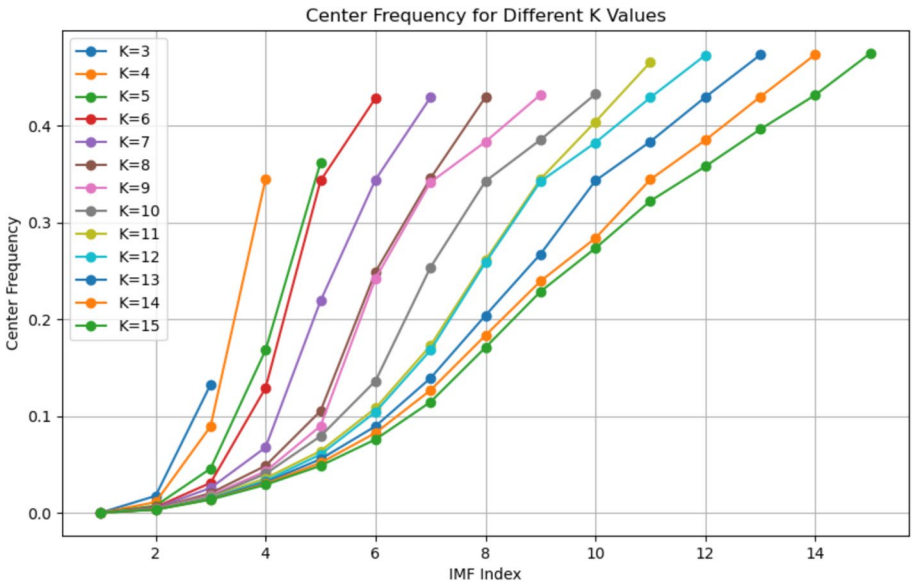


Fig. 4 Plot of center frequencies for different values of K

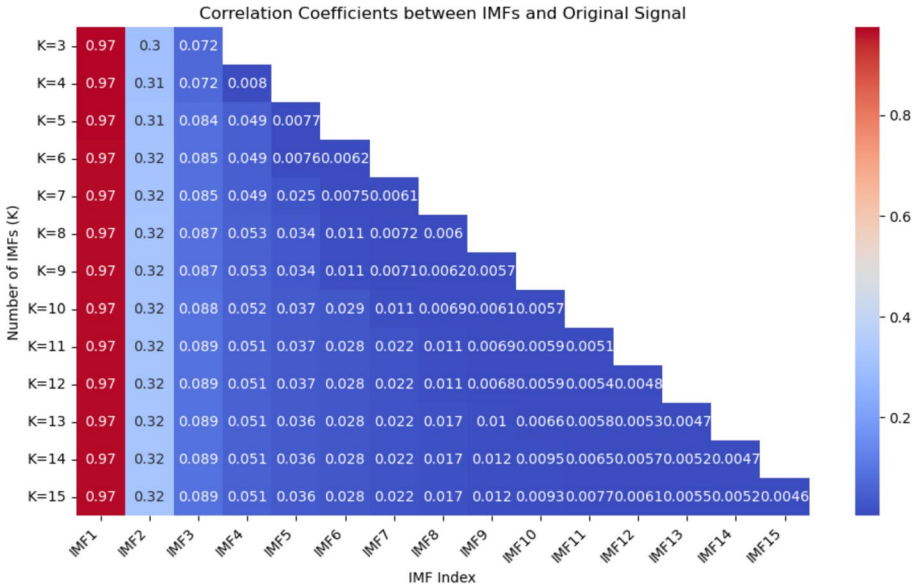


Fig. 5 Plot of correlation coefficients between IMF and original signal for different values of K

we finally choose k = 8. When the value of k is in the range of 6–10, the center frequency tends to be stable, and the coverage of the center frequency is more extensive and more uniformly distributed, which indicates that these values can better capture the different frequency components of the signal The center frequency tends to stabilize when the value of K is greater than 10. After values greater than 10, although the center frequency distribution is more detailed, it may be overfitted, i.e., contain too many unnecessary frequency components.

Finally, we decompose each of the closing prices of Shanghai Airports (sh600009) into 8 IMF components, and Fig. 6 shows the results of the VMD variational modal decomposition operation of Shanghai Airports (sh600009).

As demonstrated in Fig. 6, the IMF1 component captures the low-frequency trend in the data, reflecting the overall upward and downward trend with smooth variations. The IMF2 component reflects the lower-frequency fluctuations, showing larger periodic fluctuations. The IMF3 component captures the mid-frequency fluctuations, containing more details and oscillations. The IMF4 component shows the higher-frequency fluctuations, which are more fine-grained and contain more high-frequency noise and detail. The IMF5-IMF8 components capture higher-frequency detail and show more violent and rapid fluctuations. These components typically contain more noise and irregular fluctuations, reflecting the high-frequency component of the market.

TMFG processing with LSTM prediction

In this paper, the decomposed 8 IMF components plus 59 indicators of the original stock price data totaling 67 indicators are input into the TMFG–LSTM combination model as a sample data set. In order to improve the expressive ability of the model, the activation function is used to add nonlinear factors, and the combined model consists of an input layer, a TMFG layer, an LSTM layer, a ReLU activation layer, and an output layer.

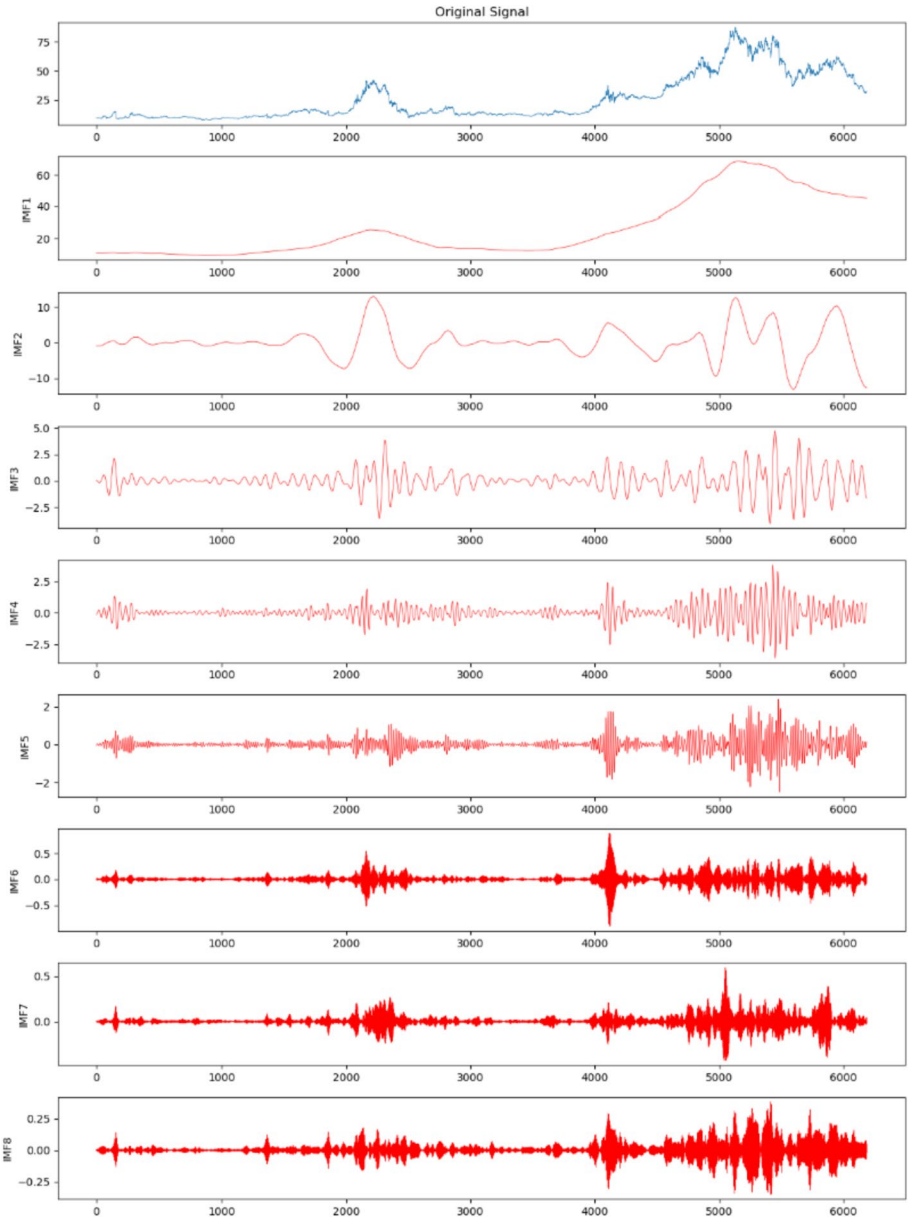


Fig. 6 VMD price decomposition result of the closing price of Shanghai Airport (sh600009) stock

Firstly, the dataset is inputted into the TMFG layer and filtered by the information filtering network to get the sparse inverse covariance matrix, then the input indicators are filtered and the original data are downsized, and then the data processed in the TMFG layer are brought into the LSTM model.

The sparse inverse covariance matrix J of the time series data of each feature is obtained by the TMFG algorithm, whose non-zero elements indicate the existence of edges (direct relationships) between the variables in the TMFG network, and zero elements indicate that the two variables are conditionally independent. Figure 7 illustrates the connectivity of the features in in the TMFG network, where connected features in the TMFG network indicate the existence of correlation, and unconnected features

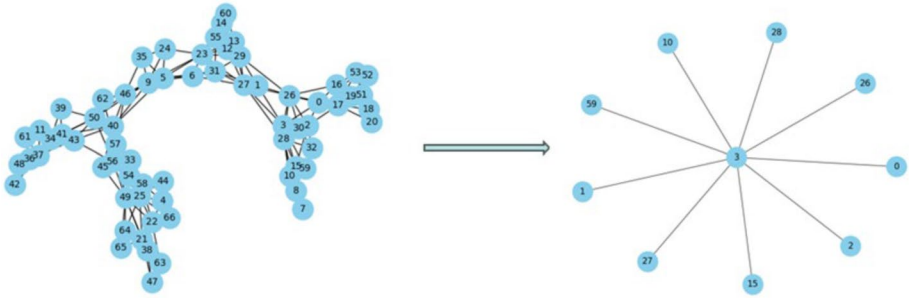


Fig. 7 TMFG decomposition results for 67 feature data of Shanghai Airport (sh600009) stock

indicate that they are conditionally independent of each other and do not have any interrelationships. After the TMFG network, we extract the nodes connected to the stock Close price (node 3) from the network. Finally, among the complex feature relationships of 67 indicator data of Shanghai Airport (sh600009) stock, we extracted Open (node 0), High (node 1), Low (node 2), Close (node 3), adjust_price (node 10), SMA_20 (node 15), PP (node 26), R1 (node 27), S1 (node 28), and IMF_1 (node 59) total 10 key features.

For the stock data of Shanghai Airports (sh600009), a total of 4955 trading days from April 22, 1998, to November 27, 2018, are selected as the training sample set, and the remaining 20%, i.e., 1237 trading days from November 28, 2018, to January 17, 2024, are selected as the test sample set. For this sample dataset, LSTM, TMFG–LSTM, VMD–LSTM, and the combined model in this paper are selected for ablation comparison experiments, and AutoRegressive Integrated Moving Average (ARIMA), Neural Network (NN), Deep Neural Network (DNN), Convolutional Neural Network (CNN) are also chosen as comparison models. The number of neurons in the hidden layers of the LSTM, VMD–LSTM, TMFG–LSTM, and VMD–TMFG–LSTM models is set to 128, with a learning rate of 0.001, the optimal number of iterations set to 100, batch_size set to 64, the number of LSTM stacked layers set to 2, and time_step set to 20.

Analysis of results

In the task of stock price prediction, the proposed VMD–TMFG–LSTM model demonstrates superior predictive performance compared to other benchmark models (ARIMA, NN, DNN, CNN, LSTM) and ablation models (VMD–LSTM, TMFG–LSTM). The ARIMA model performs poorly in capturing nonlinear relationships, with its predicted values tending to flatten, especially during periods of sharp stock price fluctuations, failing to fit the actual trends. While the CNN model shows improvement in extracting local features, its performance is still limited due to its inability to model the long-term dependencies inherent in time series data, resulting in significant prediction errors. The DNN model, although advantageous in processing high-dimensional data, similarly lacks time-series modeling capabilities, leading to suboptimal fitting results for time-series data. The LSTM model, as the baseline, can partially capture time-series trends but exhibits limited fitting capability during rapid stock price changes. Through ablation analysis, the TMFG–LSTM model effectively reduces data redundancy and noise using the TMFG algorithm, significantly improving predictive stability in highly volatile regions. On the other hand, the VMD–LSTM model incorporates variational mode

decomposition (VMD) to decompose complex time series into multiple sub-modes, enhancing the extraction of fine-grained features. Ultimately, the proposed VMD–TMFG–LSTM model integrates the strengths of both TMFG and VMD, significantly improving its ability to model nonlinear and complex time series. It achieves the lowest mean squared error (MSE) and the highest computational efficiency, maintaining outstanding predictive performance even during extreme stock price fluctuations. This demonstrates that the VMD–TMFG–LSTM model holds significant advantages in stock price prediction, offering an efficient and accurate solution for modeling nonlinear time series in financial markets. The predictive results of each model are shown in Fig. 8.

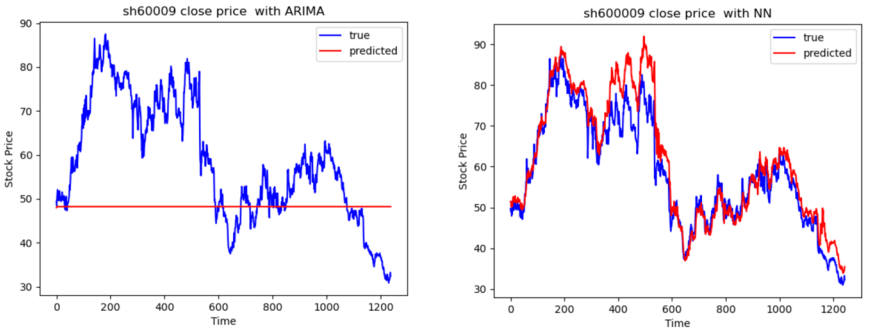
In order to improve the accuracy and credibility of the prediction comparisons among eight models, four evaluation metrics were adopted, namely Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R-Square, R^2), and Model Running Time, to assess the performance indicators of each model.

From the data in Table 3, the VMD–TMFG–LSTM model demonstrates the best performance in predicting the stock prices of Shanghai International Airport. Compared to the baseline LSTM model, the VMD–TMFG–LSTM model reduces RMSE by 69.76%, lowers MAE by 71.41%, improves R^2 by 0.2184, and reduces runtime by 46.28%. Among the ablation models, TMFG–LSTM and VMD–LSTM, the VMD–TMFG–LSTM model also outperforms them in all metrics, showcasing significant improvements in both predictive accuracy and computational efficiency. Furthermore, when compared to other benchmark models such as ARIMA (RMSE: 16.9448, R^2: −0.5880), NN (RMSE: 4.8513, R^2: 0.8697), DNN (RMSE: 3.4745, R^2: 0.9332), and CNN (RMSE: 3.6890, R^2: 0.9247), the VMD–TMFG–LSTM model consistently achieves the lowest error rates, the highest R^2, and the shortest runtime. These results highlight the effectiveness of integrating the VMD technique for signal decomposition and the TMFG algorithm for dimensionality reduction, enabling the combined model to excel in modeling complex nonlinear time series. This makes the VMD–TMFG–LSTM model a robust and efficient choice for stock price prediction tasks.

Validation of model validity

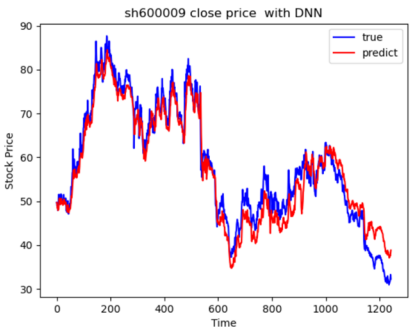
To validate the effectiveness and robustness of the proposed model, four SSE stocks were selected for experimental verification: China Resources Double-Crane Pharmaceutical Co., Ltd. (sh600062), China Meheco Group Co., Ltd. (sh600056), China Minsheng Banking Corp., Ltd. (sh600016), and Shanghai Waigaoqiao Free Trade Zone Group Co., Ltd. (sh600648). The experimental setup, including the environment, data sources, data preprocessing methods, applied models, and model parameters, is consistent with those used in the experiment for Shanghai International Airport Co., Ltd. (sh600009).

According to the results in Table 4, the VMD–TMFG–LSTM model demonstrated the best performance in stock price prediction for China Resources Double-Crane Pharmaceutical Co., Ltd. (sh600062), China Meheco Group Co., Ltd. (sh600056), China Minsheng Banking Corp., Ltd. (sh600016), and Shanghai Waigaoqiao Free Trade Zone Group Co., Ltd. (sh600648). Compared to the baseline LSTM model, the VMD–TMFG–LSTM model reduced RMSE by 49.18%, decreased MAE by 66.26%, increased R^2 by 0.0482, and shortened runtime by 58.92% for China Resources Double-Crane Pharmaceutical. For China Meheco Group, RMSE was reduced by 58.12%, MAE decreased

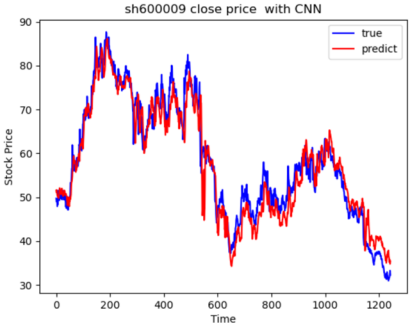


a. ARIMA prediction results

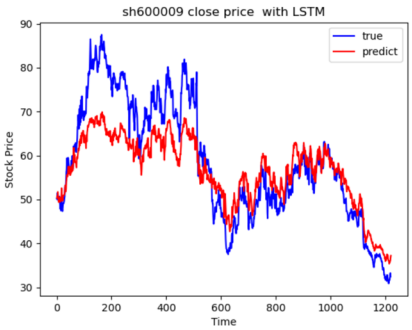
b.NN prediction results



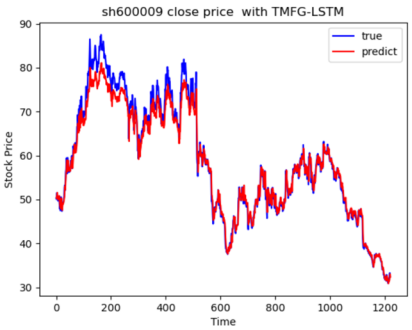
c. DNN prediction results



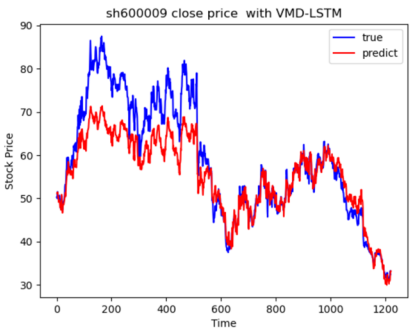
d. CNN prediction results



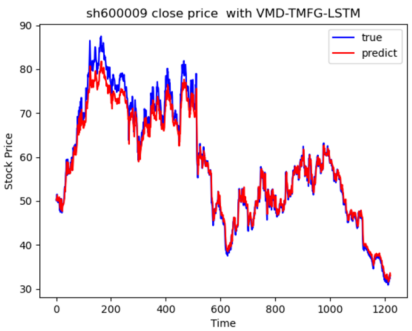
e. LSTM prediction results



f. TMFG-LSTM prediction results



g.VMD-LSTM prediction results



h.VMD-TMFG-LSTM prediction results

Fig. 8 Shanghai International Airport Co., Ltd. (sh600009) Stock Price Forecast Results

Table 3 Test set performance indicators of the eight models in Shanghai International Airport Co., Ltd. (sh600009) stock price forecasting

Stock name	Mould	RMSE	MAE	R^2	Time
Shanghai International Airport Co., Ltd. (sh600009)	ARIMA	16.9448	13.2711	−0.5880	1.50
	NN	4.8513	3.5567	0.8697	21.30
	DNN	3.4745	2.8799	0.9332	31.94
	CNN	3.6890	2.8702	0.9247	20.22
	LSTM	6.6239	5.1345	0.7596	22.74
	TMFG–LSTM	2.3046	1.6327	0.9709	16.72
	VMD–LSTM	5.6986	4.2675	0.8220	22.12
	VMD–TMFG–LSTM	2.0029	1.4677	0.9780	12.22

Table 4 Test set performance metrics for stock price prediction of four stocks

Stock name	Mould	RMSE	MAE	R^2	Time
China Resources Double-crane Pharmaceutical Co., Ltd. (sh600062)	ARIMA	4.0183	3.1929	−0.0012	1.02
	NN	1.0982	0.8955	0.9246	27.00
	DNN	0.9199	0.6248	0.9497	26.31
	CNN	1.1967	0.6378	0.9105	21.72
	LSTM	1.0254	0.8333	0.9349	32.84
	TMFG–LSTM	0.5244	0.2871	0.9829	16.05
	VMD–LSTM	0.7919	0.6374	0.9612	36.22
	VMD–TMFG–LSTM	0.5211	0.2812	0.9831	13.49
China Meheco Group Co., Ltd. (sh600056)	ARIMA	3.3305	2.1376	−0.0297	1.01
	NN	1.2487	0.9747	0.8539	19.68
	DNN	1.3300	0.8241	0.8342	15.27
	CNN	1.6235	0.8267	0.7530	21.76
	LSTM	1.3225	0.5730	0.8382	42.81
	TMFG–LSTM	0.5546	0.2858	0.9715	16.63
	VMD–LSTM	1.2382	0.5986	0.8582	37.30
	VMD–TMFG–LSTM	0.5538	0.2882	0.9716	15.26
China Minsheng Banking Corp., Ltd. (sh600016)	ARIMA	1.7660	1.5097	−2.3443	1.01
	NN	0.2945	0.2491	0.9075	17.22
	DNN	0.4862	0.4145	0.7458	22.51
	CNN	0.4303	0.3541	0.8009	35.01
	LSTM	0.3253	0.2665	0.8840	28.12
	TMFG–LSTM	0.0607	0.0425	0.9959	17.97
	VMD–LSTM	0.2724	0.2263	0.9187	24.54
	VMD–TMFG–LSTM	0.0552	0.0363	0.9966	11.91
Shanghai Waigaoqiao Free Trade Zone Group Co., Ltd. (sh600648)	ARIMA	4.8938	4.3021	−2.0641	1.02
	NN	0.6308	0.5227	0.9494	23.32
	DNN	0.7839	0.6778	0.9219	27.89
	CNN	0.7576	0.5166	0.9271	32.38
	LSTM	0.4582	0.3022	0.9730	40.78
	TMFG–LSTM	0.2870	0.1698	0.9894	22.85
	VMD–LSTM	0.2939	0.1696	0.9889	32.02
	VMD–TMFG–LSTM	0.2876	0.1708	0.9893	17.82

by 49.68%, R^2 improved by 0.1334, and runtime shortened by 64.34%. For China Minsheng Banking, RMSE dropped by 83.03%, MAE reduced by 86.41%, R^2 increased by 0.1126, and runtime shortened by 57.65%. For Shanghai Waigaoqiao, RMSE decreased by 37.23%, MAE reduced by 43.48%, R^2 improved by 0.0163, and runtime shortened by 56.32%. Overall, the VMD–TMFG–LSTM model consistently outperformed other models across all datasets, significantly reducing prediction errors, improving fitting accuracy, and enhancing computational efficiency, thereby validating its effectiveness and robustness in stock price prediction tasks.

Conclusion

To improve the stability and accuracy of stock price prediction and enhance the computational efficiency of the model, this paper integrates the VMD algorithm, the triangular maximal filtering network (TMFG) algorithm, and the LSTM network model, proposing a combined model—VMD–TMFG–LSTM. Based on closing price prediction experiments conducted on five stocks from the Shanghai Stock Exchange, the proposed model outperforms seven other models with smaller RMSE, R^2, and MAE values and the fastest running time. In complex stock price prediction tasks, the TMFG method plays a crucial role in improving the runtime and predictive accuracy of the baseline model. The VMD–TMFG–LSTM combined model presented in this paper effectively handles complex stock price data, demonstrating greater advantages compared to the other models and significantly enhancing the accuracy of stock price prediction. The following conclusions can be drawn:

(1) The introduction of TMFG algorithm to select the features of the input reduces the complexity of the input, improves the speed of iterative convergence and reduces the iterative error. The stability and robustness of prediction are improved.
(2) The VMD–TMFG–LSTM model proposed in this paper achieves the best fitting performance while ensuring model accuracy, significantly reducing runtime, and improving operational efficiency.

Since the stock closing price prediction is affected by investors' subjective factors to a certain extent, there is still room for improvement of the model in this paper. The next step is to consider introducing investor sentiment into the model to achieve better prediction results.

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Author contributions
Zhixin Zhang contributed to the conceptual design and methodology of the study. Yanrong Hu and Qingyang Liu were responsible for data analysis and algorithm implementation. Hongjiu Liu supervised the research process, with Qingyang Liu serving as the corresponding author. All authors participated in drafting, revising, and approving the final manuscript.

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Data availability
The data in this paper are available from the website yucezhe (https://yucezhe.com/).

Declarations

Competing interests
Zhixin Zhang, Qingyang Liu, Yanrong Hu and Hongjiu Liu declare that there is no conflict of interests regarding the publication of our paper "***Multi-feature stock price prediction by LSTM networks based on VMD and TMFG***".

Disclosure
Qingyang Liu, Yanrong Hu and Hongjiu Liu are the joint first authors and co-corresponding authors.

Ethics approval and consent to participate
This article does not contain any studies with human participants or animals performed by any of the authors.

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